BCSE429P

Cyber Physical Systems Design Lab

Laboratory 4 - FSM Modeling using Ptolemy II

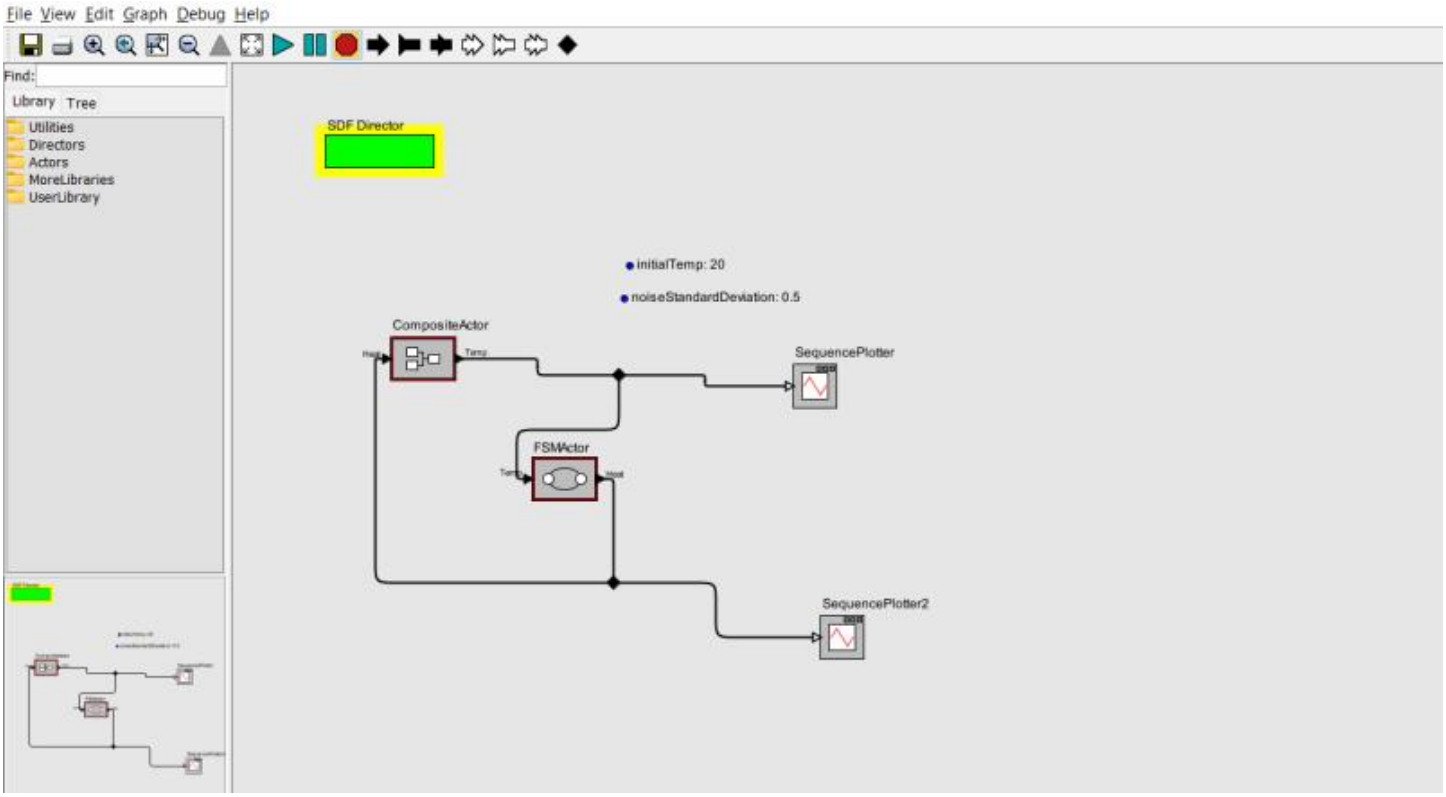
Name: Aman Kumar Singh

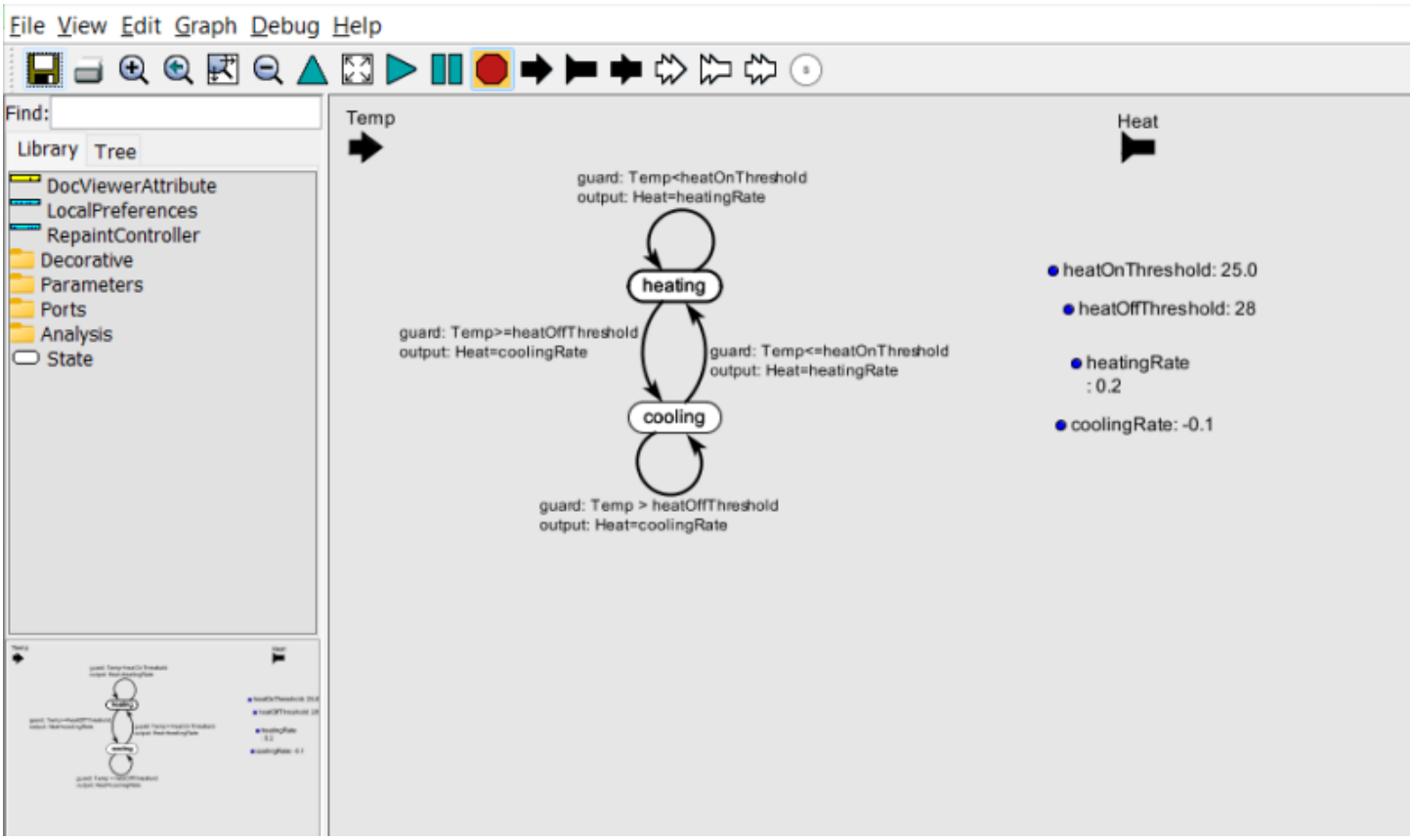
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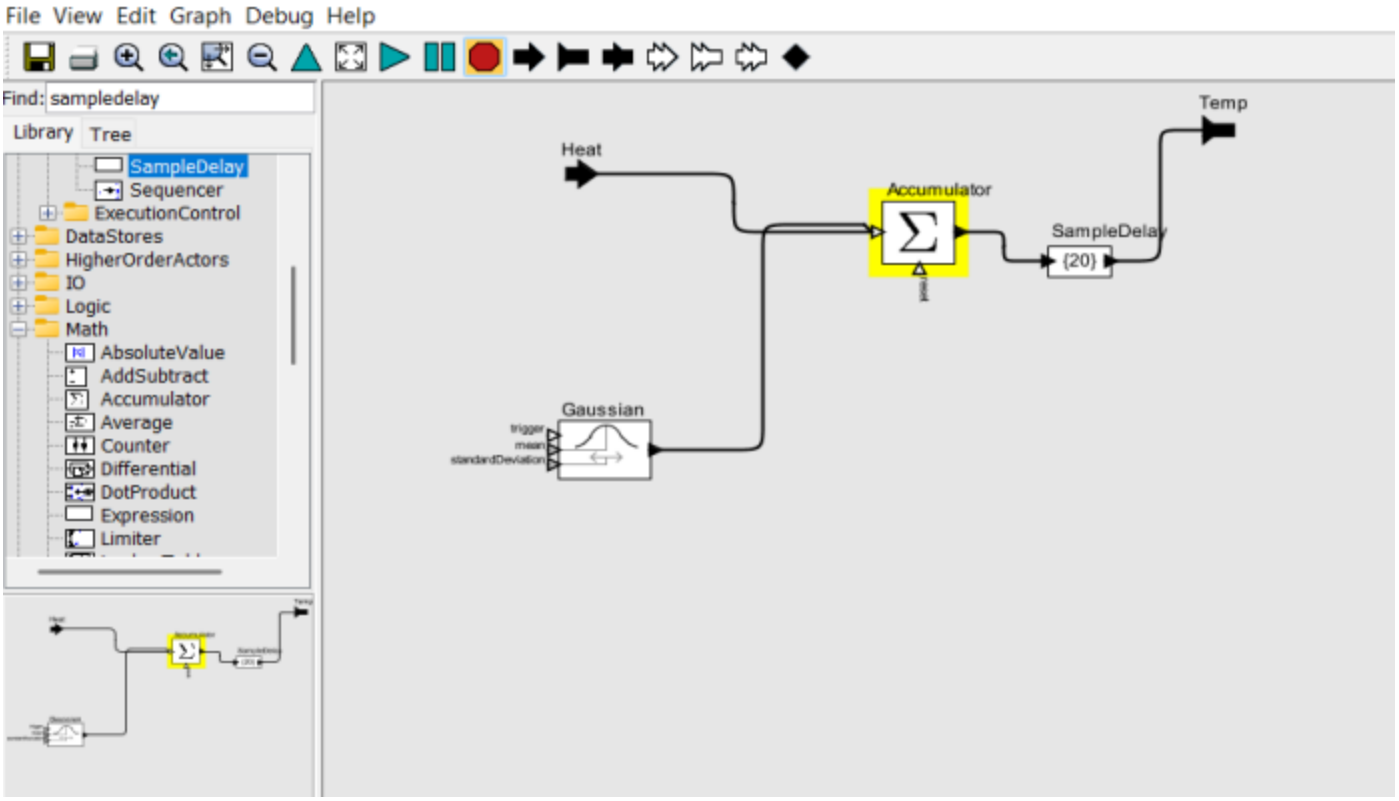
Question:

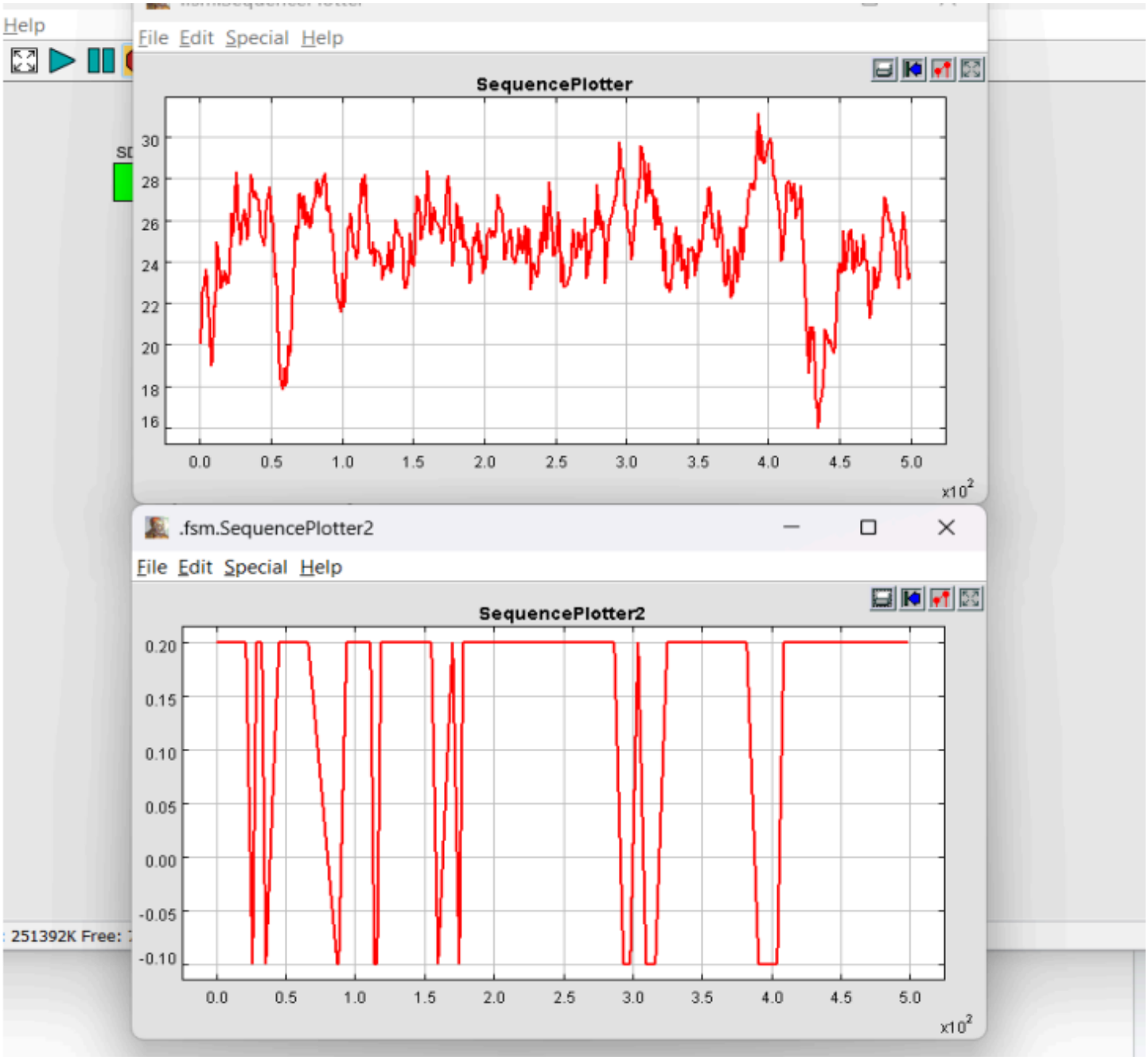
1. FSM for the Thermostat with two states: heating and cooling.
2. The heater can switch on if the temperature is less than or equal to 25 degree celcius
3. The cooler can switch on if the temperature is greater than 28 degree celcius
4. The thermostat heating rate is 0.2, and its cooling rate is 0.1.

Design an FSM model for a total iteration of 500 samples. You can use FSMactor, Composite Actor and sequence plotter to show the output.


File View Edit Graph Debug Help
Find:
Library Tree
Utilities
Directors
Actors
MoreLibraries
UserLibrary
SDF Director
initialTemp: 20
noiseStandardDeviation: 0.5
CompositeActor
SequencePlotter
FSMActor
SequencePlotter2


File View Edit Graph Debug Help
Find:
Library Tree
DocViewerAttribute
LocalPreferences
RepaintController
Decorative
Parameters
Ports
Analysis
State
Temp
Heat
guard: Temp<heatOnThreshold
output: Heat=heatingRate
heating
guard: Temp>=heatOffThreshold
output: Heat=coolingRate
guard: Temp<=heatOnThreshold
output: Heat=heatingRate
cooling
guard: Temp > heatOffThreshold
output: Heat=coolingRate
heatOnThreshold: 25.0
heatOffThreshold: 28
heatingRate
: 0.2
coolingRate: -0.1


File View Edit Graph Debug Help
Find: sampledelay
Library Tree
SampleDelay
Sequencer
ExecutionControl
DataStores
HigherOrderActors
IO
Logic
Math
AbsoluteValue
AddSubtract
Accumulator
Average
Counter
Differential
DotProduct
Expression
Limiter
Heat
Accumulator
SampleDelay
{20}
Temp
Gaussian
trigger
mean
standardDeviation


Help
File Edit Special Help
SequencePlotter
30
28
26
24
22
20
18
16
0.0 0.5 1.0 1.5 2.0 2.5 3.0 3.5 4.0 4.5 5.0
x10^2
.fsm.SequencePlotter2
File Edit Special Help
SequencePlotter2
0.20
0.15
0.10
0.05
0.00
-0.05
-0.10
0.0 0.5 1.0 1.5 2.0 2.5 3.0 3.5 4.0 4.5 5.0
x10^2
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